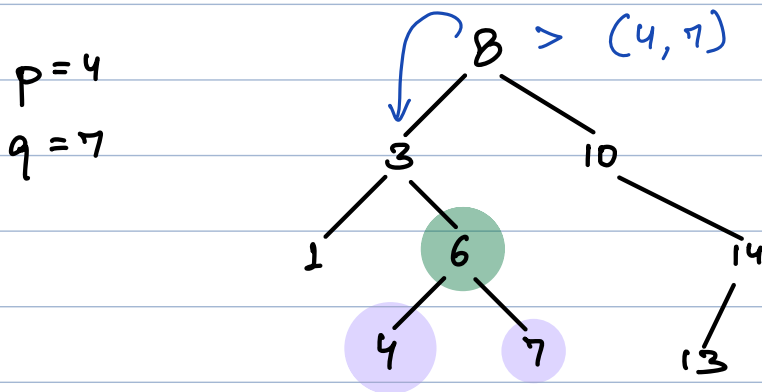
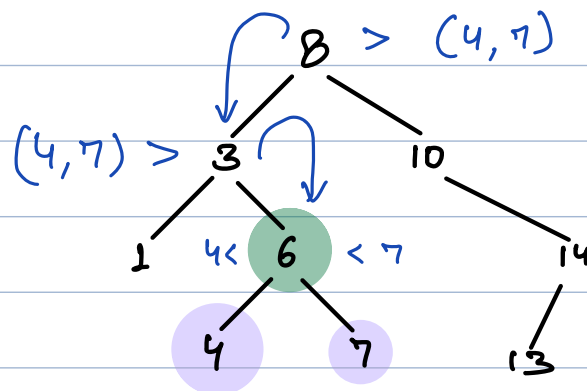


LCA in BST Given a BST A and two nodes B and C. Find the Lowest Common Ancestor.

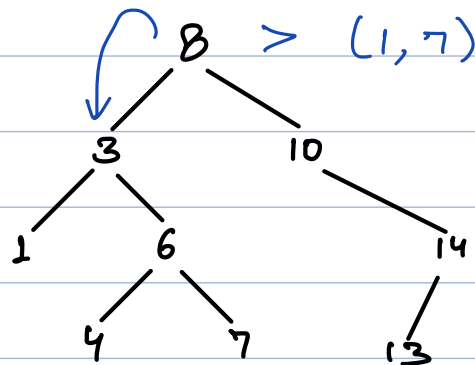


$p: 8 \ 3 \ 6 \ 4$
 $q: 8 \ 3 \ 6 \ 7$

Common: 8 3 6

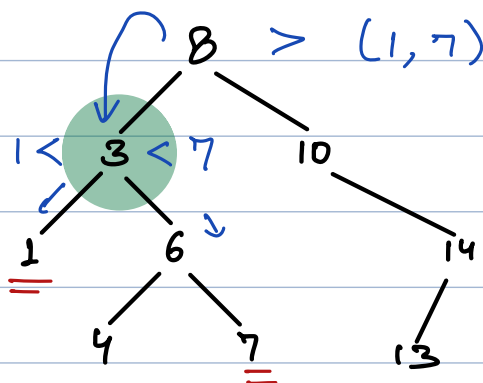


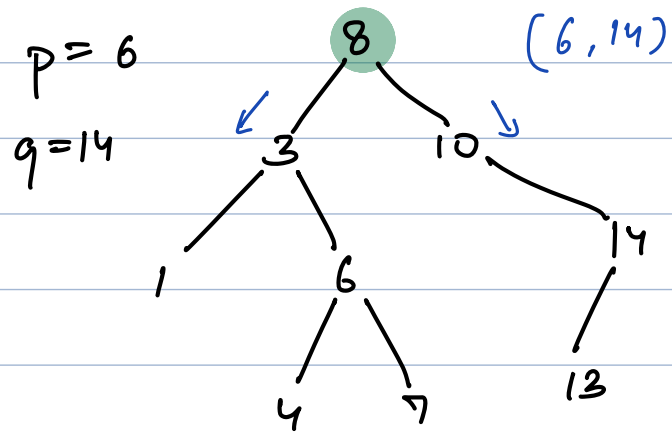
$p = 1$
 $q = 7$



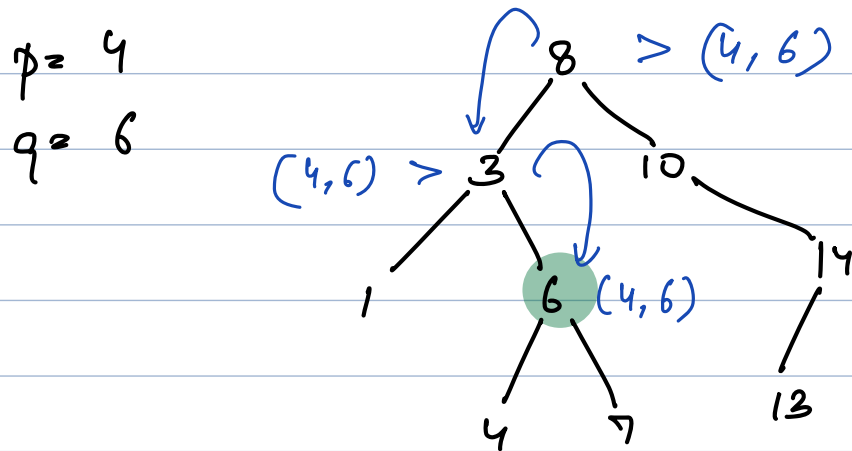
$p: 8 \ 3 \ 1$
 $q: 8 \ 3 \ 6 \ 7$

Common: 8 3





$LCA = 8$



Idea: If p and $q > \text{root} \rightarrow \text{go to right}$

If p and $q < \text{root} \rightarrow \text{go to left}$

else root is ans

```
Code
int solve(Node root, int B, int C) {
    if (root.value > B && root.value > C)
        return solve(root.left, B, C);

    if (root.value < B && root.value < C)
        return solve(root.right, B, C);

    return root.val;
}
```

TC: $O(\log N)$